AI Voice Agent Interview Platform

Project report submitted
For Review (Major Project) in fulfillment of
The requirement for the degree of

Bachelor of Technology

(Computer Science and Engineering)

By

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CERTIFICATE

This is to certify that the project report entitled “AI Voice Agent Interview Platform” submitted by “Archi Prajapati, Hitanshu Purohit, Fiza Pathan and Mahi Prajapati” to School of Engineering and Technology (SET) of Navrachana University Vadodara, in partial fulfillment for the award of the degree of B. Tech in Computer Science and Engineering (CSE) department. This report is a bona fide record work that has been carried out under my supervision during academic year 2025-26. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

Dr. Avani Vasant	**Prof. Ninad Bhavsar**	**Dr. Ashish Jani**
Guide and Assistant Professor	Program Chair and Assistant Professor	Head and Professor
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School of Engineering and Technology	School of Engineering and Technology	School of Engineering and Technology
Navrachana University, Vadodara	Navrachana University, Vadodara	Navrachana University, Vadodara
May 2025	May 2025	May 2025

Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

In today's highly competitive job market, communication, confidence, and interview preparedness play an equally vital role as technical expertise. However, most candidates lack access to realistic, voice-interactive tools that replicate the pressure and dynamics of an actual interview. The AI Voice Agent Interview Platform addresses this challenge by leveraging Artificial Intelligence (AI), Natural Language Processing (NLP), and Speech Recognition to simulate real-time, human-like interviews.

The system enables candidates to engage with an AI-powered interviewer that asks contextually relevant questions, listens to verbal responses, and evaluates them based on clarity, tone, and content. It provides instant, personalized feedback to help users enhance their communication and performance. For organizations and academic institutions, it offers an efficient, unbiased, and scalable approach to preliminary screening and candidate assessment.

Built with a modular architecture integrating AI-driven question generation, voice interaction APIs, and real-time analytics, the platform represents a significant step toward merging technology with career development. Ultimately, the AI Voice Agent Interview Platform redefines how individuals prepare for interviews making practice more intelligent, interactive, and accessible.

Keywords: AI Interview Platform, Natural Language Processing (NLP), Voice-Based Interview, Mock Interview Simulation, Candidate Assessment

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Project Title:

AI Voice Agent Interview Platform

1. Scope

The scope of this project is to develop an AI-powered voice-based interview platform capable of simulating real interview scenarios. The system will enable candidates to interact with an AI agent that asks questions verbally, listens to responses, and provides feedback or evaluation based on speech content, tone, and confidence level. The platform will integrate technologies such as Natural Language Processing (NLP), Speech Recognition, and Machine Learning to create a realistic interview experience. It can be used by students for interview preparation, by training institutes for skill evaluation, and by organizations for preliminary screening of candidates.

Problem Statement and Objective:

In the existing recruitment and interview training process, candidates often lack access to interactive, voice-based practice platforms that simulate real interviews. Most available tools are text-driven and fail to replicate the pressure, communication flow, and response dynamics of actual interview sessions. This leads to inadequate preparation, communication anxiety, and inefficient candidate evaluation by recruiters.

To address this problem, the proposed AI Voice Agent Interview Platform aims to automate and enhance the interview experience using AI and voice technologies. The objective is to create an intelligent system that conducts automated voice interviews, analyzes user responses, and provides instant feedback. The platform will not only assist recruiters in reducing manual screening efforts but will also help candidates improve their communication, confidence, and performance through personalized analysis and practice sessions.

2. Motivation

In today's competitive job market, effective communication and interview skills are crucial for career success. However, many candidates struggle to gain confidence and experience before facing real interview situations. Traditional mock interviews often require significant time, human involvement, and scheduling effort, making them inaccessible to many students and job seekers. Additionally, organizations spend a large amount of time and resources conducting initial screening rounds for multiple applicants, many of which could be automated.

With the rapid advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), it has become possible to replicate human-like conversations and interactions through intelligent systems. This inspired the idea of developing an **AI Voice Agent Interview Platform** that can simulate a human interviewer, ask contextually relevant questions, listen to candidate responses, and evaluate their performance automatically.

The motivation behind this project is to bridge the gap between theoretical knowledge and practical communication skills by providing an easily accessible, voice-based AI system for interview practice and screening. Such a solution can empower candidates to prepare more effectively, reduce anxiety, and build confidence, while also assisting recruiters and educators in evaluating candidates more efficiently.

3. Literature Review

Sr. No.	Platform	Merits	Demerits
1	**Grow with Google: Interview Warmup**	A free tool that helps users practice interview questions and analyze spoken answers with transcriptions and insights.	Does not generate AI-based questions; lacks interactive and adaptive interview features.
2	**AIApply**	Generates tailored interview questions based on uploaded job descriptions and provides instant answer evaluations.	Focused mainly on question generation; minimal conversational or adaptive interaction.
3	**Interviewsby.ai**	Offers mock interviews customized to specific job descriptions with instant AI-generated feedback and improvement tips.	Limited free trials; evaluation lacks deep speech and emotion analysis.
4	**SmallTalk2Me**	Analyzes over 30 parameters including fluency, grammar, vocabulary, and confidence—ideal for non-native speakers.	Emphasizes language and fluency rather than behavioral or technical interview preparation.
5	**Interview Sidekick**	Functions as a real-time AI assistant during interviews, providing contextual guidance and access to a large question bank.	Does not fully simulate interviewer–candidate interaction; limited natural voice feedback.
6	**InterviewAI (App)**	Mobile app offering voice-based mock interviews with instant feedback and personalized question sets.	Limited analytical depth and lacks adaptive follow-up questioning.
7	**AI Interview IQ (App)**	Conducts real-time voice-to-voice mock interviews tailored to various industries, improving communication and presentation skills.	Lacks strong focus on technical or coding interviews; may not provide detailed evaluation reports.

8	**The AI Interview (App)**	Records user responses and generates personalized performance feedback.	Does not include dynamic question flow or adaptive conversation.

Table 3.1 Literature Review

4. System Requirements for Development and Production Environment

The successful development and deployment of the AI Voice Agent Interview Platform demands a set of robust system requirements covering both hardware and software aspects, ensuring reliable operation during model development, voice processing, and deployment.

1. Hardware Requirements

- Processor: Minimum Intel Core i5 (10th Gen) or equivalent; higher recommended for faster training and response times.
- RAM: At least 8 GB; 16 GB preferred, especially for handling multiple interviews or training deep learning models.
- Storage: SSD with at least 256 GB; large datasets or audio files require additional space.
- Graphics: Discrete GPU (e.g., Nvidia GTX 1650 or above) for AI/ML acceleration and real-time voice processing.
- Audio: High-quality USB or condenser microphone for clear voice input; speakers or headphones for audio output.
- Network: Stable broadband internet for API integrations, cloud deployment, and candidate access.

2. Software Requirements

- Operating System: Windows 10/11, Ubuntu 20.04+, or MacOS (latest stable version).
- Programming Languages: Python 3.9+ (NLP, backend), JavaScript (Node.js for backend services, React/Next.js for frontend).
- Core Libraries & Frameworks:
    - TensorFlow/PyTorch (AI model development)
    - Hugging Face Transformers (NLP)
    - SpeechRecognition, PyAudio, Vapi.ai (voice processing)
    - Flask or FastAPI (backend server)
    - React.js, Next.js, TailwindCSS, Shadcn/UI (frontend)

    - Zod (data validation).

- Database: Firebase, Supabase, or MongoDB for authentication, user data, and feedback reports.

- Cloud & APIs: Google Gemini or OpenAI API for AI responses; Google Cloud/AWS for deployment; Vapi for real-time voice interaction.

- DevOps Tools: Git & GitHub for version control, Docker for containerization, Visual Studio Code/PyCharm as IDEs.

3. Production Environment

- The platform is deployed on a scalable cloud infrastructure: AWS EC2, Google Cloud Run, or Vercel for the web client.

- CI/CD pipelines are set up using GitHub Actions and Docker to enable safe, continuous updates.

- Role-based access controls and encryption are employed to ensure security of interview data and feedback.

- Real-time monitoring and logging via cloud dashboards facilitate robust maintenance and troubleshooting after deployment.

5. Stakeholders

The AI Voice Agent Interview Platform is designed for a diverse group of stakeholders spanning users, developers, organizations, and technology partners. Each group interacts with the system in unique ways and contributes to its development, deployment, and real-world impact.

1. Students and Job Seekers

- Use the platform to practice technical and HR interviews in a simulated environment and receive instant feedback to refine their interview skills.
- Gain confidence and improve communication, technical knowledge, and interview strategies before participating in real job interviews.

2. Academic Institutions and Training Centers

- Integrate the system into placement preparation modules, professional development courses, and feedback-driven learning programs for student upskilling.
- Analyze candidate performance and measure improvement using automated reports and analytics.

3. Recruiters and HR Professionals

- Employ AI-driven interviews for candidate pre-screening, saving considerable time and standardizing initial evaluation processes.
- Review feedback, communication ratings, and candidate reports to make informed hiring decisions before personal interviews.

4. Platform Developers and Engineers

- Build, integrate, and maintain the software’s backend, frontend UI/UX, voice processing modules, and AI feedback system.
- Ensure system reliability, efficient data management, and seamless cloud deployment of the platform.

5. Technology Partners and API Providers

- Supply essential tools, APIs, and frameworks for voice processing (Vapi.ai), database management (Firebase/Supabase), AI modeling (Google Gemini, OpenAI, Hugging Face), and web development (React, Next.js, TailwindCSS, Shadcn/UI).

6. Mentors and Academic Guides

- Monitor student progress, evaluate project milestones, and ensure ethical standards are maintained during platform development and deployment.
- Provide guidance and feedback for continuous improvement of the system's usability and reliability.

7. Organizational End Users

- Use the platform for internal recruitment, workforce skill assessment, and HR process automation in companies and organizations.
- Benefit from data-driven AI interview insights and streamlined candidate evaluation workflows.

6. Approach Used

The AI Voice Agent Interview Platform has been developed using a systematic, modular, and user-centric approach. This ensures the solution is robust, scalable, and meets the diverse requirements of all stakeholders involved.

1. Requirement Analysis

- Defined clear objectives: Automate mock interviews, provide feedback, and prepare candidates for real interviews.
- Identified critical features based on stakeholder needs such as voice interaction, instant analytics, feedback breakdown, and retake options.

2. Technology Stack Selection

- Selected modern tools for efficient development, including Next.js, React.js, TailwindCSS, and Shadcn/UI for responsive and scalable frontend interfaces.
- Used Python and AI APIs for natural language processing and interview feedback analysis.
- Integrated real-time voice processing using Vapi.ai for conversational AI interviews, and managed data/authentication with Firebase and Supabase.

3. Modular Development & Integration

- Developed UI and backend modules in parallel, focusing on clean interfaces for question selection, interview flow, feedback report generation, and dashboard management.
- Implemented APIs for voice agent communication, question generation, user authentication, and data persistence using modular functions.

4. Iterative Testing and Feedback

- Adopted Agile methodology for continuous development and client feedback cycles, ensuring rapid prototyping and feature improvements.
- Validated conversational accuracy, technical skill assessment, and system usability through user testing and mentor evaluation.

5. Deployment and Enhancement

- Deployed on cloud platforms (AWS, Vercel, or Google Cloud Run) for scalable access and seamless performance.
- Incorporated end-user feedback for future enhancement planning: support for additional job roles, languages, detailed analytics, and improved voice agent intelligence.

7. Challenges Faced and Solutions Implemented

During the development and deployment of the AI Voice Agent Interview Platform, various challenges were encountered at technical, operational, and user-experience levels. This section discusses the significant hurdles and the solutions adopted to overcome them.

1. **Voice Recognition Accuracy**

- **Challenge:** Accurate transcription of interviewee's spoken responses was difficult due to variations in accent, speech clarity, and background noise.
- **Solution:** Integrated state-of-the-art voice recognition APIs from Vapi.ai and utilized noise cancellation and audio pre-processing techniques to improve speech clarity and recognition accuracy.

2. **Real-Time Response and Latency**

- **Challenge:** Ensuring seamless real-time conversational flow was hindered by network delays and API response times.
- **Solution:** Optimized backend processing by asynchronous API calls, implemented caching where possible, and ensured low-latency cloud deployment on regionally proximate servers.

3. **Dynamic Question Generation and Relevance**

- **Challenge:** Automatically generating and selecting interview questions relevant to the candidate’s job profile and interview type was complex.
- **Solution:** Developed metadata tagging of questions based on difficulty levels, domain expertise, and question types, and used contextual AI prompts to tailor conversations dynamically.

4. **Feedback Generation and User Understanding**

- **Challenge:** Translating raw interview data into meaningful, understandable feedback for candidates posed user experience difficulties.
- **Solution:** Designed structured feedback reports with clear scoring metrics, categorized skill assessments, and personalized recommendations for improvement.

5. **Integration and Data Security**

- **Challenge:** Integrating diverse APIs and ensuring data security, especially personal and interview data, was critical and complex.
- **Solution:** Adopted role-based access control, encrypted data storage, secure API gateways, and compliance with privacy regulations to protect user data.

8. Data Corpus Used

The quality and relevance of the interview questions and feedback generated by the AI Voice Agent Interview Platform heavily depend on the underlying data corpus. Careful curation and selection of this dataset ensure the platform can simulate realistic interviews across technical, behavioral, and aptitude domains.

1. Source of Data

- Publicly Available Interview Question Sets:
  Curated from common technical interview question repositories, coding challenge databases, and behavioral question collections to cover a wide range of industries and job roles.
- AI Generated Content:
  Leveraged natural language generation models and prompt engineering techniques to create new, relevant, and context-sensitive questions.
- Domain-Specific Content:
  Included questions specific to roles such as full stack development, front-end, back-end, and other fields to tailor the interview experience.

2. Types of Data

- Technical questions covering programming languages, algorithms, data structures, and system design principles.
- Behavioral questions assessing communication skills, problem-solving, teamwork, leadership, and cultural fit.
- Aptitude questions involving logical reasoning, quantitative analysis, and analytical thinking.

3. Data Preprocessing

- Standardized question formats and answers to facilitate AI comprehension and delivery.
- Filtered out ambiguous or outdated questions to maintain quality and relevance.
- Annotated questions with metadata such as difficulty level, domain, and question type for dynamic interview generation.

9. Data Flow Diagram / UML

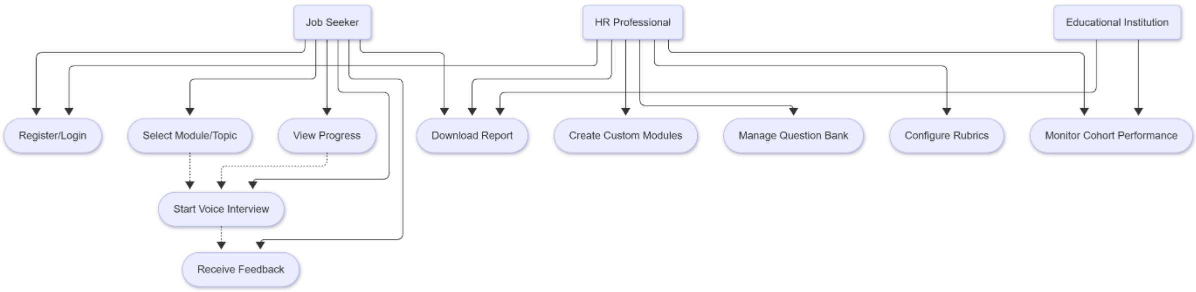

Figure 9.1 Use Case

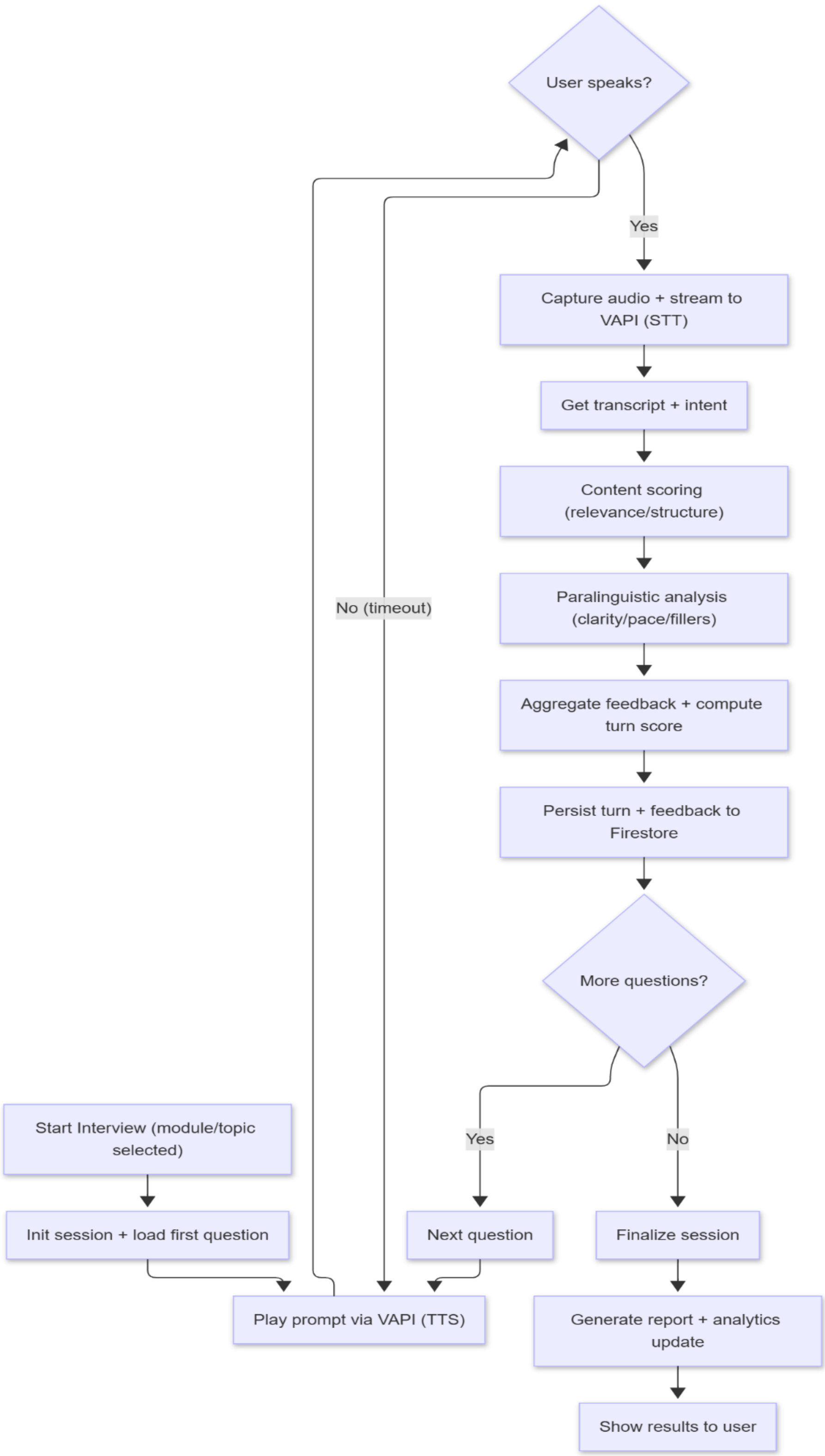

Figure 9.2 Activity Diagram

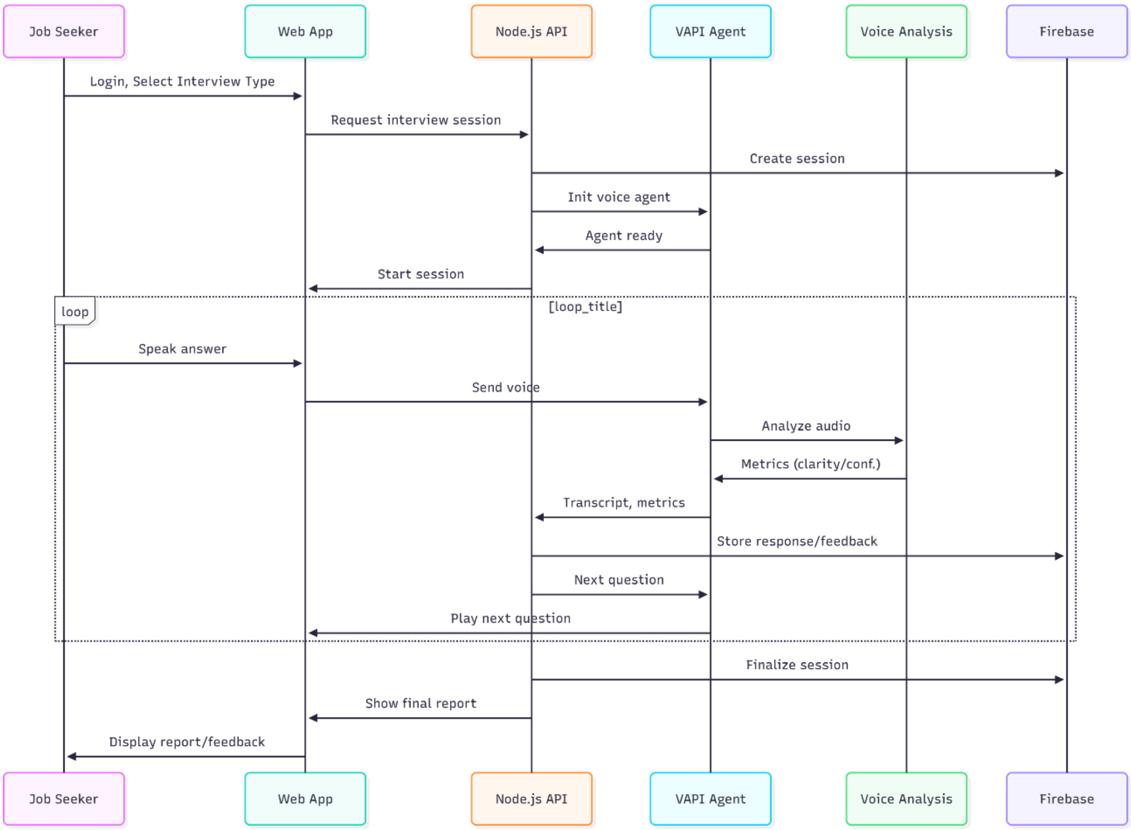

Figure 9.3 Sequence Diagram

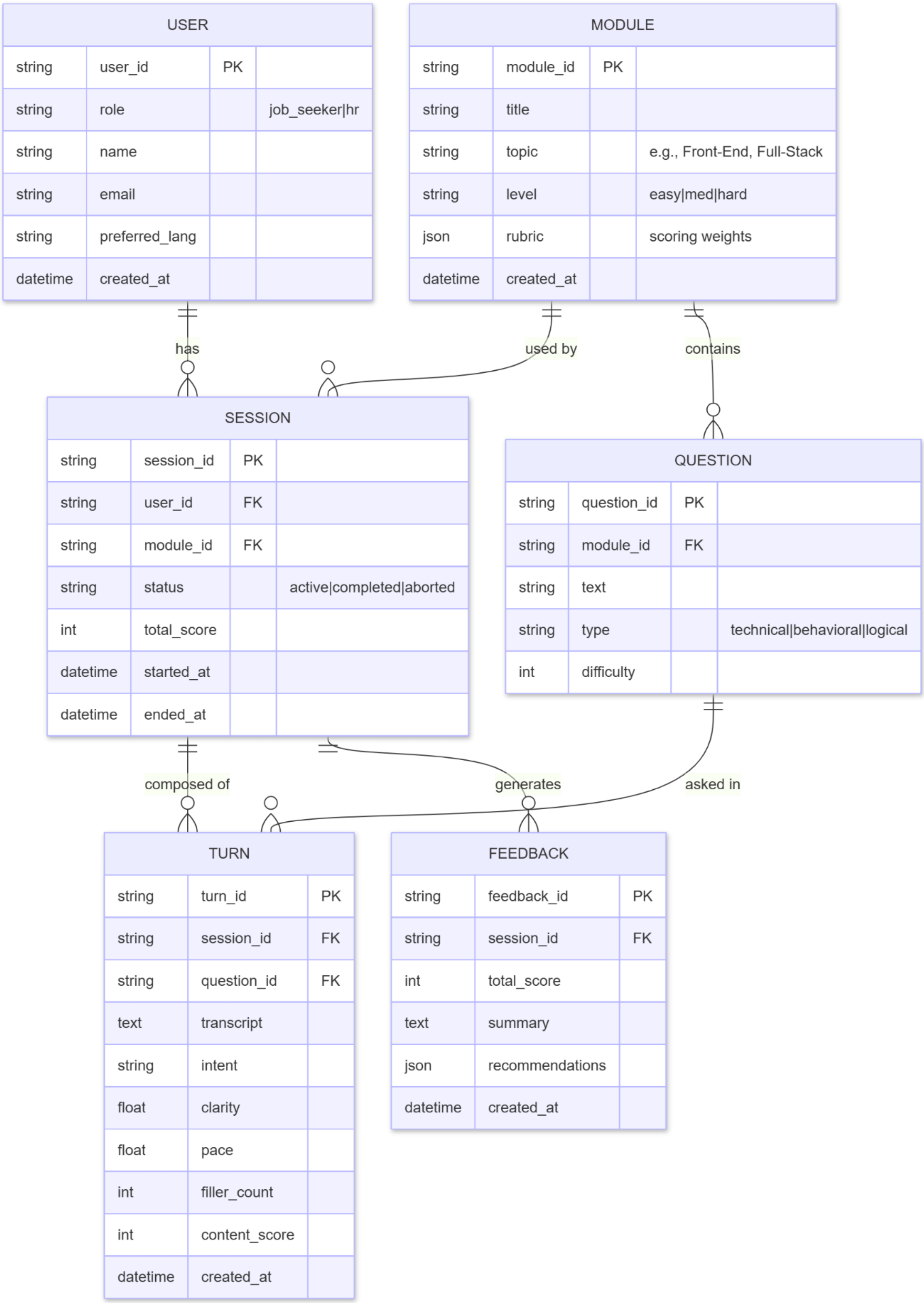

Figure 9.4 ER Diagram

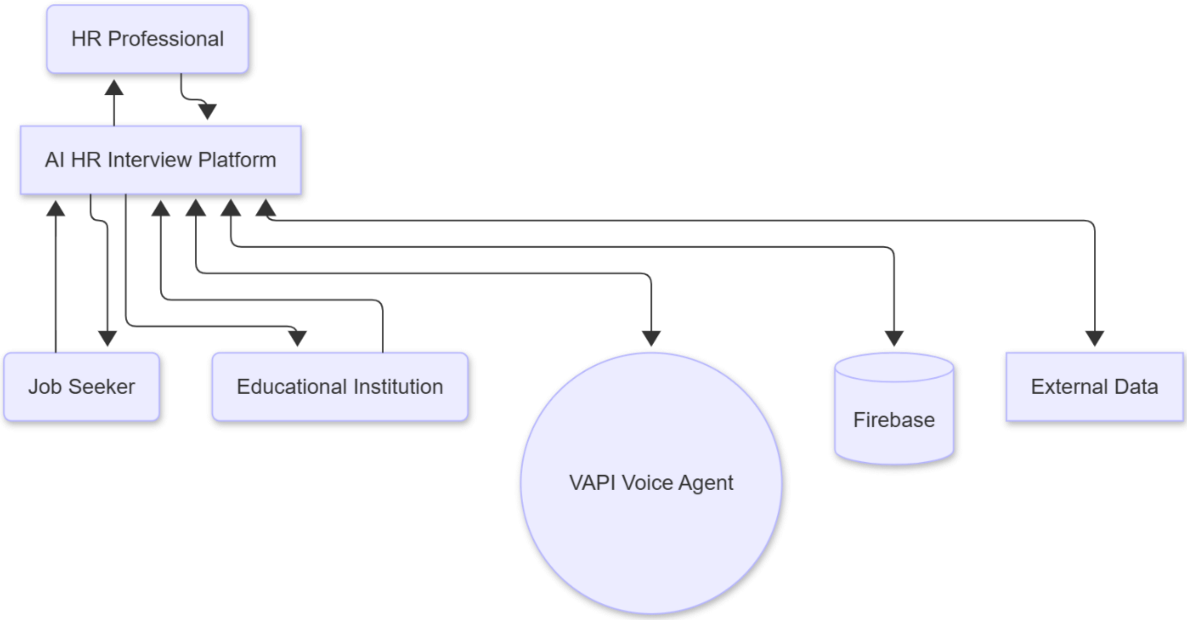

Figure 9.5 DFD Level – 0

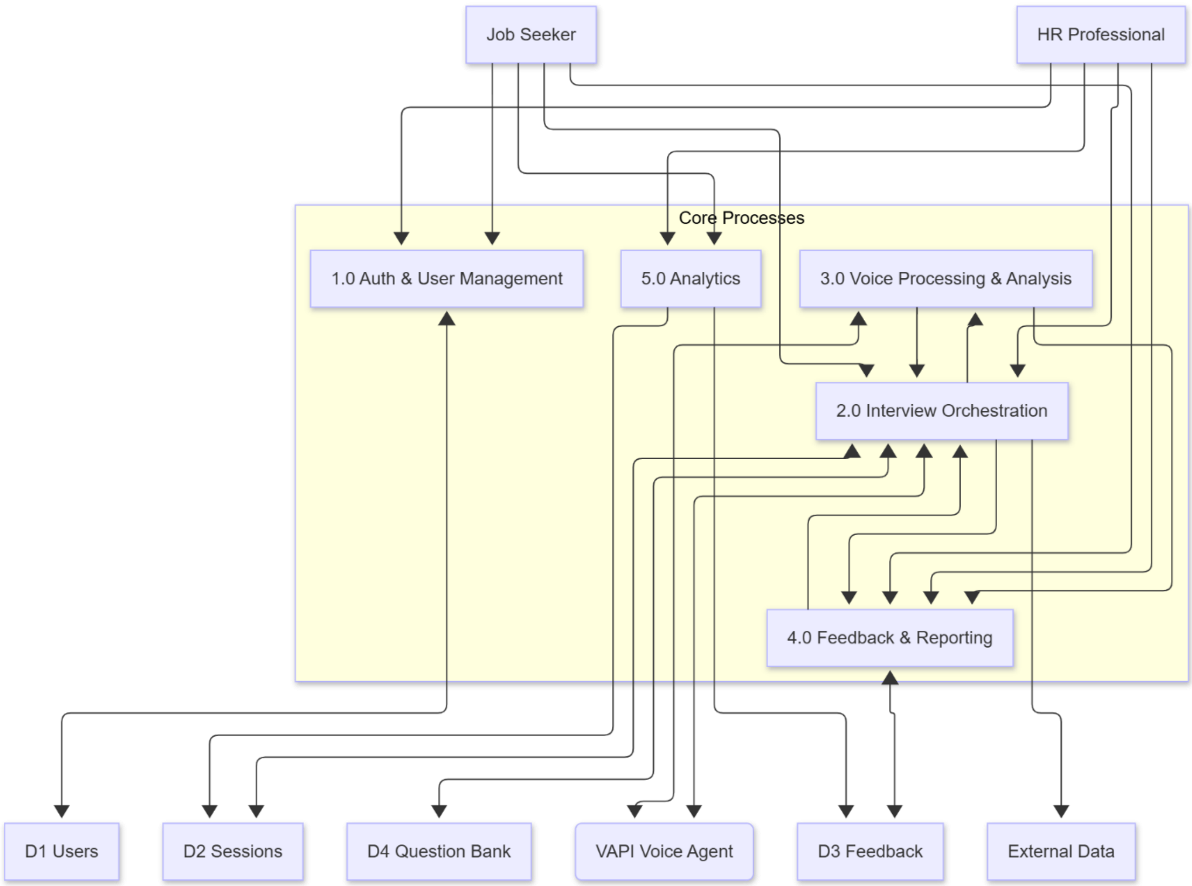

Figure 9.6 DFD Level – 1

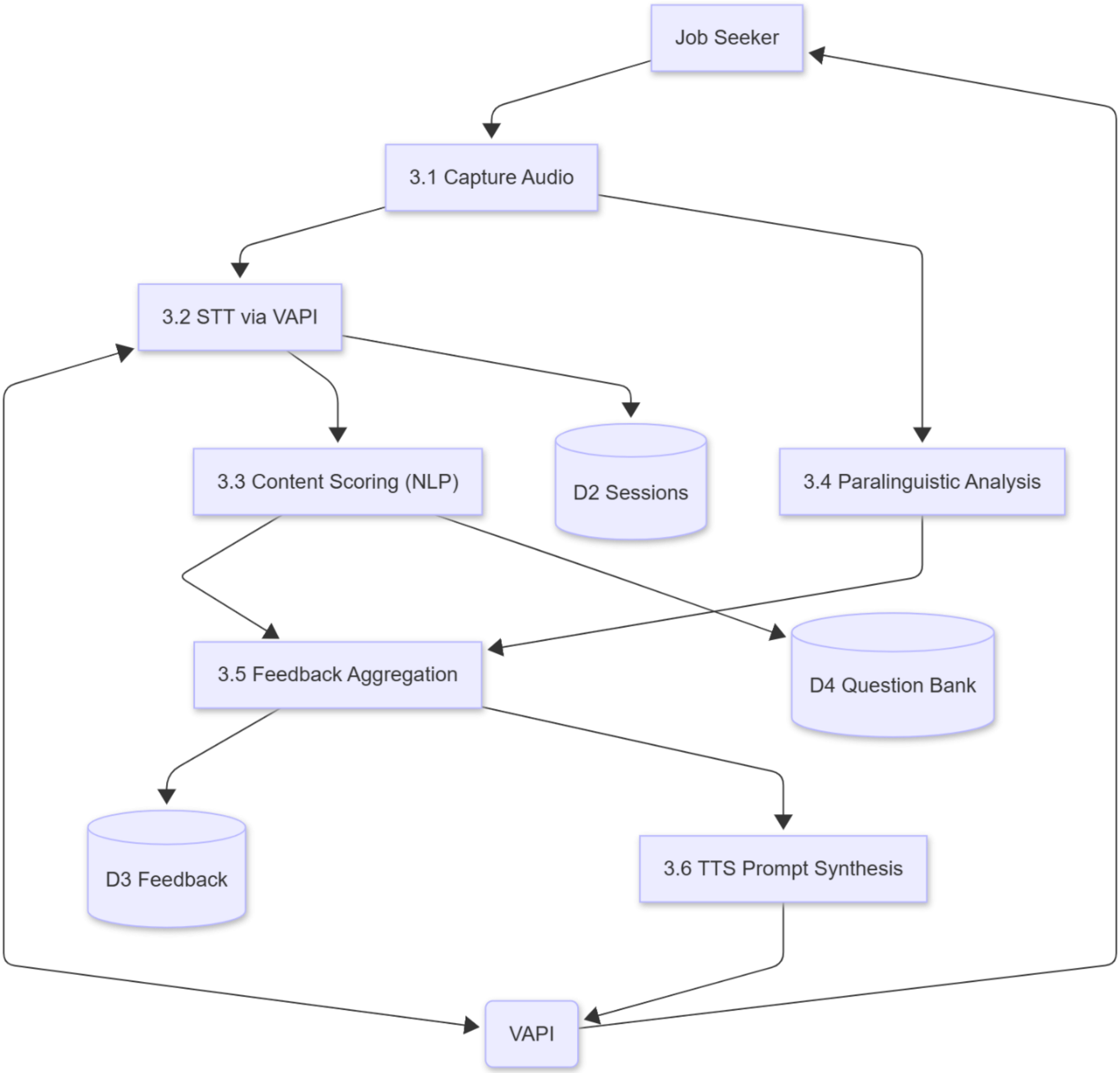

Figure 9.7 DFD Level – 2

10.Prototype Implementation’s Screenshot

Sharing link because the quality of the prototype design will be not good in Screenshot – https://app.eraser.io/workspace/UPvCvAJeMHVWB950WKNU?origin=share

11.Gantt Chart / Timeline

Target 1: Project Initiation and Scope Definition (September 2024)

- Finalized topic: *AI Voice Agent Interview Platform*.
- Conducted initial brainstorming on problem statement, goals, and expected outcomes.
- Defined project scope, objectives, and initial features (e.g., voice-based interviews, real-time feedback).

Target 2: Data Collection and System Design Planning (October 2024)

- Gathered datasets and question corpora for various interview types (technical and behavioral).
- Outlined system architecture and workflow diagrams (frontend–backend–API–database).
- Scheduled work division among team members for development activities.

Target 3: Frontend Prototype Development (November 2024)

- Began development of the frontend prototype using Next.js, React, and Tailwind CSS.
- Implemented UI components like login, dashboard, and interview setup screens.
- Integrated Firebase authentication for user and admin access control.

Target 4: Backend Implementation and API Integration (December 2024 – January 2025)

- Developed backend logic for question generation, AI-driven feedback, and voice communication using Vapi.ai and Google Gemini APIs.
- Connected the frontend with APIs to ensure real-time interview flow.
- Structured the database using Firebase/Supabase to store user, interview, and feedback information securely.

Target 5: Feature Enhancement, Testing, and Deployment (February – May 2025)

- Added advanced modules: candidate analytics, result visualization, and retake options.
- Improved user interface and overall application performance based on mentor feedback.
- Conducted detailed testing (unit, functional, and user acceptance).
- Completed documentation, report writing, and deployment of the web-based system.

12.Proposed Enhancement

The **AI Voice Agent Interview Platform** is currently in its early development phase, with core functionalities such as voice-based interview simulation, question generation, and basic performance feedback already planned or partially implemented. As the project progresses, several enhancements have been identified to further improve usability, intelligence, and accessibility in future iterations.
The following are the key proposed enhancements:

1. **Improved Voice Interaction Quality**

   Future updates will focus on making the AI interviewer's voice more natural and context-aware. By refining speech synthesis and emotion detection, the system will respond with better tone variation and conversational smoothness, creating a more realistic interview experience.

2. **Expanded Question Database**

   A larger and more diverse set of technical, HR, and behavioral questions will be incorporated. This enhancement aims to cover additional job domains and varying difficulty levels, ensuring a wider application across industries and user needs.

3. **Offline Practice Mode**

   A lightweight offline mode is proposed for candidates with limited internet access. This will allow users to attempt practice interviews locally and later sync data once reconnected.

4. **Integration with Resume Analysis Module**

   A future version may include a resume parsing feature that extracts key skills and generates personalized interview questions accordingly. This will help align candidate responses with their actual profiles.

13.Conclusion

The **AI Voice Agent Interview Platform** is an innovative step toward modernizing the interview preparation and evaluation process through the use of Artificial Intelligence and voice interaction technologies. The project aims to simulate real-time interview scenarios where candidates can interact with an AI interviewer, receive feedback, and improve their communication and confidence levels in a convenient, accessible environment.

Through this platform, students and job seekers can practice interviews without human intervention, making the preparation process more scalable and efficient. Recruiters and academic institutions can also benefit from automated candidate screening, ensuring consistent and unbiased assessments.

Since the project commenced in **September 2025**, the team has focused on developing the core modules including voice-based interaction, question generation, and AI-driven feedback. While the system is still under active development, the initial phase has successfully laid the foundation for a functional prototype that demonstrates the feasibility and potential impact of the platform.

In the upcoming phases, the project will focus on integrating advanced enhancements that will further enhance the system’s performance and usability, making it a valuable tool for both interviewees and organizations.

Overall, the project contributes to bridging the gap between theoretical learning and practical communication skills, showcasing how AI can empower individuals to become more confident, well-prepared, and job-ready in today’s competitive world.

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